

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	30-06-2026
Team ID	23A51A4211
Project Name	Visitor Management System
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Visitor Pre-Registration	• Select Host Employee Name • Enter Visitor First & Last Name • Enter Visitor Corporate Email • Enter Visitor Contact Phone Number • Select Expected Arrival Date & Time
FR-2	Dynamic Input Validation	• Validate Phone Number String Layout Rules • Enforce Mandatory Field Rule Completions • Prevent Form Submission on Regex Pattern Violations
FR-3	Log Request Submission	• Submit Visitor Entry Log Record • Generate Unique Platform Sys_ID Tracking Key • Instantiate Custom Log Row in x_snc_visitor_log Table
FR-4	Lobby Check-In Processing	• Display Front Desk Security Desk Kiosk Form • Provide One-Click 'Check-In' Interactive Button • Capture Real-Time Check-In Timestamp Metadata
FR-5	Notification Management	• Send Automated Arrival Confirmation Email to Host Employee • Notify Security Dispatchers on Unexpected Walk-in Records • Dispatch Transactional App Status Notifications
FR-6	Digital Badge Identification	• Generate Secure Virtual Visitor Pass Layout • Generate Quick Response (QR) Verification Key Code • Attach PNG QR Badge Asset to Active Log Record

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability / Experience	The system must provide an intuitive, simple, self-service user portal allowing business hosts or external visitors to input logistical details within 3 steps without requiring prior software training.
NFR-2	Platform Security	The application must restrict access utilizing row-level Access Control Lists (ACLs) and assigned system roles, ensuring sensitive logging rows are encrypted and viewable only by authorized security administrators over secure HTTPS connections.
NFR-3	Data Validation Reliability	The system script execution engine must achieve 99.9% accuracy when evaluating dynamic validation constraints, formatting inputs accurately before allowing persistence transactions.
NFR-4	Transactional Performance	Server processing, validation logic checks, state transitions via UI Actions, and QR code asset generations must execute and refresh within 2 seconds under peak load parameters.
NFR-5	Operational Availability	The scoped cloud components must inherit native cloud infrastructure availability vectors, maintaining continuous 24x7 operation with minimal downtime for building entrance check-ins.
NFR-6	Architectural Scalability	The application data model must seamlessly scale to manage thousands of concurrent visitor registrations and active log entries without degrading execution speeds or interface responsive rates.